

The Long-term Impact of the Historical Slave Trade in Africa on Contemporary Industrialization^{*}

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Abstract

This study seeks to elucidate one of the channels through which the historical slave trade in Africa contributed to the region's current economic underperformance. By integrating contemporary individual-level occupational data with the volume of slave exports during the four major slave trades from 1400 to 1900, I identify a negative relationship between the historical slave trade and contemporary industrialization. Through rigorous controls for confounding factors and applying an instrumental variable strategy that uses the distances from each country to the primary destinations of the slave trades, I establish that this relationship is causal. Furthermore, by analyzing the differential adverse impact of the historical slave trade on contemporary industrialization across countries with varying degrees of pre-colonial institutional centralization, I find that the adverse impact of the slave trades on contemporary industrialization was sustained by the generally poor institutional quality of African governments.

JEL Codes: C36, N47, N67, O14, O43, O55

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1 Introduction

The historical slave trade has been identified as a significant factor contributing to Africa's current economic underperformance (Nunn, 2008). Building on Nunn's pioneering work, a substantial body of research has explored the long-term effects of the historical slave trade on various other outcomes (as will be discussed in the literature review below). However, there remains a gap in understanding the channels through which the historical slave trade led to current economic underperformance except Nunn and Wantchekon (2011), which suggests mistrust as a channel through which the historical slave trade persistently affects the region's current economic performance.

Given the widely accepted view that industrialization can significantly spur economic growth (Solow, 1956; Kaldor, 1966, 1967; Kuznets, 1973; Kogan et al., 2017) and the observation that African countries still lag in contemporary industrialization (Diao et al., 2017; Nguimkeu and Zeufack, 2024), this paper investigates whether the historical slave trade impeded Africa's economic development by hindering its industrialization process. Moreover, the slave trades disproportionately captured more men than women (Manning, 1990; MacGaffey, 2000). Although recent literature argues that the female workforce expanded during the process of industrialization (Goldin and Sokoloff, 1982), men nevertheless played a larger role at the onset of industrialization (Pinchbeck, 1930; Horrell and Humphries, 1995). This further suggests that the historical slave trade may be linked to patterns of industrialization.

I begin by introducing the qualitative historical context relevant to the subjects of analysis. First, I highlight the emergence of a vicious cycle in African societies during the historical slave trade, illustrating how it likely impeded collective efforts toward industrialization. Second, I discuss the early stages of industrialization in African countries after they gained independence in the mid-twentieth century, which were characterized by significant government intervention and sowed the seeds of subsequent deterioration in institutional quality (Bates, 2014, 2015). Lastly, I explore the history of institutional development in Africa, with a focus on the relationship between pre-colonial institutional centralization and current institutional quality (Gennaioli and Rainer, 2007).

Motivated by the qualitative historical evidence that the historical slave trade might hinder the process of contemporary industrialization in Africa, I investigate the impact of the historical slave trade on contemporary industrialization quantitatively. In practice, I employ country-level data on

the volume of the historical slave trade from [Nunn \(2008\)](#) and the contemporary share of workers in industrial sectors from [WorldBank \(2023\)](#). Additionally, I analyze individual-level occupational data from the 2016 (Round 6) Afrobarometer survey ([Afrobarometer, 2016](#)), which are nationally representative surveys encompassing 36 African countries. The results from country-level regressions reveal a negative correlation between the volume of the historical slave trade and the contemporary share of workers in industrial sectors. This finding is further supported by individual-level regressions, which show a negative correlation between the volume of the historical slave trade and the probability of respondents currently being employed in industrial sectors.

Following the methodology outlined in [Altonji et al. \(2005\)](#), I compute the extent to which the selection on unobservable factors must exceed that of observable factors in order to fully explain away my estimates. The calculation indicates that the impact of unobservable factors would need to be 2.5 to 4.5 times larger than that of observable factors to nullify the observed OLS negative relationship. Given that I incorporate multiple sets of meticulous controls in both country-level and individual-level regressions, it is unlikely that my OLS estimates suffer from such significant bias.

I carefully control for individual characteristics in the Afrobarometer survey, as well as country-level geographical and climatic factors that could bias the estimates, but relying solely on OLS results is still insufficient to establish a causal relationship. Therefore, I employ an instrumental variable (IV) approach, using the distances from each country to the primary destinations of each of the four major slave trades as instruments. The instrumental variable estimates reinforce the OLS findings, showing significantly negative coefficients of interest with a slightly larger magnitude than the OLS estimates. The controls and the instrumental variable strategy provide robust evidence that the historical slave trade causally hampered the process of contemporary industrialization in Africa.

After establishing the causality of the negative relationship between the historical slave trade and contemporary industrialization, I examine how the impact of the slave trades on industrialization persisted over centuries. Historical qualitative evidence indicates that the trajectory of contemporary industrialization in Africa has been markedly influenced by significant government intervention ([Rweyemamu, 1979](#); [Steel and Evans, 1981](#)). In light of this, I focus on the role of institutional quality as the primary factor perpetuating the adverse impact of the historical slave trade on industrialization. This focus is also supported by existing literature emphasizing the importance of

institutional quality as a key determinant in facilitating industrialization ([Acemoglu et al., 2002](#)). Considering that government dysfunction and corruption are critical obstacles to economic development in Africa ([Bates, 2014, 2015](#)), I hypothesize that the negative impact of the historical slave trade on contemporary industrialization is likely to be more pronounced in countries with lower institutional quality.

Directly comparing the impact of the historical slave trade across countries with differing levels of institutional quality today is problematic, as current institutional quality may be endogenous, potentially shaped by various factors during the colonial and post-colonial periods that are possibly correlated with the slave trades. To address this issue, I estimate the impact of the historical slave trade on contemporary industrialization separately for countries with high or low degrees of pre-colonial institutional centralization, which is associated with current institutional quality ([Gennaioli and Rainer, 2007](#)) but not a consequence of colonial and post-colonial developments as it predates them. The findings reveal that the adverse impact of the historical slave trade on contemporary industrialization is more pronounced in countries with lower degree of pre-colonial institutional centralization, suggesting that institutional factors played a crucial role in sustaining or even exacerbating the long-term adverse impact of the historical slave trade on industrialization.

Contributions to the Literature

To the best of my knowledge, this paper is the first to formally explore the relationship between the historical slave trade and contemporary industrialization in Africa. By doing so, I provide a possible channel through which the historical slave trade contributes to current economic underperformance in Africa documented in [Nunn \(2008\)](#). My work is related to several strands of literatures.

First, this paper contributes to the broader understanding of the long-term effects of the historical slave trade in Africa. As discussed earlier in this section, following [Nunn \(2008\)](#), numerous studies have examined various long-term consequences of the historical slave trade, such as ethnic fractionalization ([Whatley and Gillezeau, 2011](#); [Green, 2013](#); [Whatley, 2014](#)), political violence, ethnic conflicts, national identity ([Zhang et al., 2021](#)), trust ([Nunn and Wantchekon, 2011](#)), female labor force participation, female genital cutting, gender norms ([Teso, 2019](#); [Corno et al., 2025](#)), matrilineal kinship ([Lowes and Nunn, 2024](#)), and polygyny ([Edlund and Ku, 2011](#); [Dalton and Leung,](#)

2014; Bertocchi and Dimico, 2019). Moreover, Bertocchi and Dimico (2019) establishes a link between the prevalence of polygyny, driven by slave trades, and rising HIV rates. However, the exact channels through which the historical slave trade led to current economic underperformance remain relatively understudied. This paper makes a significant contribution by not only identifying an additional long-term outcome of the historical slave trade but also by establishing a key channel—industrialization—through which the historical slave stagnated economic growth in Africa.

Secondly, my work contributes to the ongoing exploration of the process and timing of industrialization in non-Western countries. For instance, Amsden (1991) investigates the divergent industrialization trajectories in East Asia, with particular attention to why the Philippines lagged behind other East Asian countries such as Japan, Korea, and Taiwan, despite the adoption of extensive export promotion policies. Similarly, Hsieh (2002) explores the industrialization process in East Asia using a novel approach to estimate total factor productivity growth (TFPG). In Latin America, Baer (1972) evaluates the outcomes of the Import Substitution Industrialization (ISI) policy, the dominant strategy employed to foster industrialization and economic growth in the region, which ultimately failed and, in some cases, even delayed industrial development. Further comparisons between East Asia and Latin America by Jenkins (1991) suggest that greater state autonomy in East Asia facilitated its rapid industrialization. In post-independence India, ISI was also initially implemented as a strategy to promote industrial growth, later transitioning toward more export-oriented policies (Kuruvilla, 1996). Additionally, Bell and Rousseau (2001) highlights the critical role that the development of financial intermediaries played in advancing industrialization in India.

In the context of Africa, using Kenya as an illustrative case, Ksoll et al. (2021) argues that violent conflicts, particularly during election periods, disrupted production and impeded sustained industrial growth. Moreover, McMillan and Zeufack (2022) and Rodrik et al. (2022) emphasize that the dominance of small, informal firms with low productivity in African manufacturing sectors recent years posed a significant barrier to industrialization and economic progress.¹ My paper offers a novel perspective: the historical slave trade also played a critical role in hampering the process of industrialization in Africa.

Third, I contribute to the literature on the role of institutions in shaping economic growth. Early

¹ As will be further discussed in Subsection 7.1, even larger firms with large-scale establishments often lack efficiency and, in some cases, prove redundant.

research highlights the importance of institutions that protect private property in fostering economic development ([DeLong and Shleifer, 1993](#); [Jones, 2003](#)). Building on this foundation, [Acemoglu et al. \(2001b, 2002\)](#) demonstrate that former European colonies experienced stronger economic growth when institutions emphasizing the protection of private property and checks on governmental power were established, as evidenced by the experiences of the New World countries.² In contrast, in regions where extractive institutions were implemented—particularly in most African countries—income levels remained stagnant. Furthermore, [Acemoglu and Johnson \(2005\)](#) finds that institutions focusing on the protection of property rights (property rights institutions) have a more substantial positive impact on the economy compared to those focusing on enabling private contracts between citizens (contracting institutions). Institutions also play a crucial role in the context of financial markets. For instance, [Acemoglu et al. \(2003\)](#) demonstrates that extractive institutions are more responsible for economic volatility and crises than macroeconomic policies. Similarly, [Beck et al. \(2003\)](#) finds that institutions protecting private property rights facilitated better development of financial intermediaries.

The closest works to my paper are [Michalopoulos and Papaioannou \(2013, 2014\)](#), which present a strong association between the degree of pre-colonial institutional centralization and current regional development. I extend this body of literature by emphasizing the role of centralized pre-colonial institutions, which is associated with higher subsequent institutional quality ([Gennaioli and Rainer, 2007](#)), in safeguarding the process of contemporary industrialization, even in the face of significant adverse external shocks such as the historical slave trade.

Lastly, this research is related to the literature on the long-run effects of historical conflicts. On the one hand, civil wars, in particular, have been shown to exert negative persistent effects on economic development ([Collier et al., 2009](#)). For instance, [Cassar et al. \(2013\)](#) utilizes variation in exposure to civil wars in Tajikistan to assess how historical conflicts undermine impersonal exchanges by eroding interpersonal trust. Similarly, [Rohner et al. \(2013\)](#) examines regional differences in the intensity of ethnic conflicts in Uganda, demonstrating that post-conflict recovery was slower in areas with higher ethnic fractionalization. On the other hand, [Bauer et al. \(2016\)](#) reviews wars across the globe and suggests that exposure to war-related violence tends to foster more cooperative and

²Specifically, the New World countries here refer to Australia, Canada, New Zealand, and the United States.

altruistic behaviors, contributing to some seemingly miraculous post-war economic and social recoveries. In a broader context, [Guiso et al. \(2009\)](#) investigates the relationship between trust—often shaped by historical conflicts—and international trade in Europe, finding a generally positive association.

There is also literature addressing the legacy of pre-colonial historical conflicts in Africa. Notably, [Besley and Reynal-Querol \(2014\)](#) finds that the presence of pre-colonial historical conflicts is associated with a greater prevalence of post-colonial conflicts and economic underdevelopment in Africa. My research contributes to this body of work by introducing industrialization as an additional economic consequence of historical conflicts, thus deepening our understanding of how the legacy of historical conflicts continues to shape current economic development.

The remainder of the paper is structured as follows. The next section details the historical backgrounds, providing qualitative foundation to the subjects of my analysis. Section 3 describes the data sources, sample construction, and presents descriptive statistics. Section 4 displays two stylized graphs: one explores the negative correlation between the volume of the historical slave trade and contemporary industrialization, while the other shows the relationship between the distances from each country to the primary destinations of the slave trades, my instrumental variables, and the volume of the historical slave trade. Section 5 outlines the empirical strategy employed at various levels of analysis. Section 6 presents the main findings, demonstrating that the observed negative correlation between the historical slave trade and contemporary industrialization is causal, while also discussing the validity of the identification strategy. Section 7 investigates the institutional mechanism through which the adverse impact of the historical slave trade on industrialization persisted over centuries. Section 8 concludes the paper.

2 Historical Background

In this section, I detail the historical background of the slave trades, the early stages of industrialization, and institutional development in Africa, providing qualitative evidence in support of the main quantitative analysis in this paper.

2.1 The Historical Slave Trade in Africa

The historical slave trade in Africa comprised four major slave trades: the trans-Atlantic, trans-Saharan, Indian Ocean, and Red Sea slave trades, spanning from 1400 to 1900. Among them, the trans-Atlantic slave trade was the largest and most well-known, involving the shipment of slaves from Central Africa (including West-Central and East-Central Africa) to the New World countries³, which were European colonies at the time. Approximately 12 million slaves were exported in the trans-Atlantic slave trade, a figure that far exceeds the combined total of the other three slave trades, which amounted to roughly half that number ([Manning, 1990](#)). In particular, men were disproportionately captured during the slave trades ([Manning, 1990](#); [MacGaffey, 2000](#)), resulting in skewed sex ratios.

Through the slave trades, a vicious cycle emerged. In a climate of pervasive insecurity brought about by the slave trades, Africans were compelled to seek weapons for self-defense against enslavement. These weapons were often obtained from Europeans, who exchanged them for slaves. Consequently, Africans armed with guns would enslave others to protect themselves from being enslaved, further exacerbating the region's insecurity ([Alpers, 1969](#); [Isaacman, 1989](#); [Hawthorne, 1999](#); [Klein, 2001](#); [Lovejoy, 2011](#)). Given the existence of such a brutal cycle, it is difficult to envision how societies with this history could later unite to foster industrialization, which inherently require much cooperation and stably effective institutions.

2.2 The Early Stages of Industrialization in Africa

Prior to independence, industrialization in Africa was limited to few selected industries, such as food processing, textiles, and wood products, primarily to serve the interests of colonial powers ([Ewing, 1968](#)). Infrastructure, taking railways as an example, were also constructed primarily to meet the military and imperial demands of the colonizers ([Kamarck, 1971](#)). As a result, by the time African countries gained independence, they generally lacked the necessary conditions for industrialization, including capital, human resources, and infrastructure. That said, even after achieving political independence, African economies remained heavily dependent on foreign exports, technologies, and capital ([Roemer, 1981](#)). To illustrate the extent of African countries' reliance on foreign economies

³Here, the New World countries again refer to Australia, Canada, New Zealand, and the United States.

at the time of their independence, it is notable that even in the most industrialized nations in the region, such as Zimbabwe, Kenya, and Uganda, there were no domestically owned establishments with more than 10 employees in the 1960s, the time they gained independence ([Acharya, 1981](#)).

Therefore, at that time, government intervention was widely accepted as essential for promoting industrial development. Consequently, many African nations adopted import substitution policies, which involved raising tariffs and implementing protective trade policies ([Rweyemamu, 1979](#)). These policies initially indeed spurred growth in industrial sectors ([Steel and Evans, 1981](#)). However, the persistence of government intervention, often characterized by corruption and inefficiency, has since become a barrier to contemporary industrialization ([Bates, 2014, 2015](#)), as will be elaborated further in Subsection 7.1.

2.3 Institutional Development in Africa

As proposed in [Michalopoulos and Papaioannou \(2013, 2014\)](#), pre-colonial ethnic institutions played a crucial role in shaping subsequent regional development. There was considerable heterogeneity in the degree of political centralization across pre-colonial African ethnic groups, ranging from small village chiefdoms to large centralized states with formal administrative structures and hierarchical organizations, as documented by [Murdock \(1959, 1967\)](#).

One might suggest that pre-colonial ethnic institutions were drastically altered by European colonial powers. However, contrary to this suspicion, European colonial rule imposed relatively limited changes on the structure of these pre-colonial institutions, largely due to the constraints of both the duration and the scope of colonial control ([Davidson, 1993](#); [Herbst, 2014](#); [Mamdani, 2018](#)). As a result, current institutional quality remains influenced by pre-colonial institutions, as emphasized by [Gennaioli and Rainer \(2007\)](#), which also offers a way to address the endogeneity of present-day institutional quality potentially shaped by various factors during the colonial and post-colonial periods that are correlated with the slave trades, as will be discussed further in Subsection 7.2. Moreover, the literature shows that the quality of pre-colonial institutions—shaped by factors such as broad tribal participation ([Acemoglu et al., 2001a](#)) or the natural environment ([Fenske, 2013](#); [Alsan, 2015](#))—continues to influence institutional structures and economic development today.

Nevertheless, it is important to acknowledge a parallel body of literature that explores the vary-

ing styles of European colonial rule in Africa and the underlying factors driving these differences. [Acemoglu et al. \(2001b\)](#) argues that the mortality rates of European settlers, affected by the local malaria ecology, shaped the type of institutions established. In areas with high settler mortality, extractive institutions were implemented, while regions with lower mortality saw the establishment of institutions that protected property rights. These differences in colonial institutions were shown to have a significant and lasting impact on regional disparities in current income levels ([Acemoglu et al., 2002](#)). However, this paper primarily utilizes the relationship between the degree of pre-colonial institutional centralization and current institutional quality, as highlighted by [Gennaioli and Rainer \(2007\)](#), rather than the institutional differences introduced during the colonial period because the main objective of my analysis is the interaction between the historical slave trade and institutional quality instead of the broader colonial experience.

3 Data

I combine several sources of data to evaluate the impact of the historical slave trade on contemporary industrialization and investigate how the impact persisted several centuries. In this section, I describe data sources, sample construction, and present descriptive statistics.

3.1 Data Sources

Slave Exports I utilize country-level historical slave export data for 52 African nations, as provided by [Nunn \(2008\)](#), which estimates the volumes associated with the four major slave trades: the trans-Atlantic, trans-Saharan, Indian Ocean, and Red Sea slave trades, spanning four historical periods (1400-1599, 1600-1699, 1700-1799, 1800-1900). This dataset also includes information on the colonial power prior to African countries' independence, as well as geographical variables such as absolute latitude (distance from the equator), longitude, and proximity to the ocean (measured by the natural log of coastline divided by land area). Additionally, climatic variables are collected, including minimum monthly rainfall, maximum monthly afternoon average humidity, and the minimum average monthly lowest temperature. These variables are incorporated as controls to mitigate potential selection in regions that were disproportionately targeted during the historical slave trade.

Distances to the Primary Destinations of the Slave Trades Nunn (2008) also provides the distances from each country to the primary destinations of the four major slave trades, which are employed as instrumental variables in my analysis. Specifically, in the trans-Atlantic and the Indian Ocean slave trades, distance is measured as the sailing distance from the point on the coast nearest to a country's centroid to the closest major slave-importing market. In the trans-Saharan and the Red Sea slave trades, distance is measured as the overland distance from a country's centroid to the nearest slave-exporting port.

Contemporary Industrialization Levels To assess contemporary country-level industrialization, I utilize data from WorldBank (2023). This dataset offers various indicators related to industrialization, with the share of employment in industrial sectors in the year 2000 being the primary measure used in this study. The choice of the year 2000 ensures consistency and comparability with the results from Nunn (2008), which uses GDP in the year 2000 as a proxy of current economic performance.

For the individual-level analysis, I utilize the 2016 (Round 6) Afrobarometer survey (Afrobarometer, 2016), which includes responses from 36 African countries, the largest number of surveyed countries across all Afrobarometer surveys, with a sample size of 53,935 respondents. The survey asks respondents to provide information about their current occupation or their previous occupation if they are currently unemployed. I categorize these responses into agricultural, industrial, or service sectors.⁴ Respondents with missing values for occupation, as well as those reporting "never had a job" or "student," are excluded from the analysis. The Afrobarometer surveys also collect basic individual characteristics such as age, gender, urban or rural residency, education level, religion, and current living conditions. These variables allow for precise control and reduce potential omitted variable bias in the analysis.

Pre-colonial Institutional Centralization To investigate the institutional quality mechanism that persisted the adverse impact of the historical slave trade on industrialization, I utilize the "jurisdictional hierarchy beyond the local community level" index from Murdock (1967). This index quantifies the degree of centralization among pre-colonial ethnic institutions in Africa. The data were collected from a series of field studies conducted in the late nineteenth and early twentieth centuries, thereby predating developments during the colonial and post-colonial periods. Following the lit-

⁴See detailed occupations in Appendix C.

erature (Alsan, 2015), I classify ethnic groups with values of 1 (petty chiefdoms) or 2 (paramount chiefdoms) as having low pre-colonial institutional centralization, while those with values of 3 or 4 (large states) are classified as having high pre-colonial institutional centralization. The original variable in Murdock (1967) is defined for each ethnic group within a given country. For the purposes of my country-based analysis, I adopt the maximum score across ethnic groups in each country as a proxy for the degree of pre-colonial institutional centralization at the country level.

3.2 Sample Construction

In the country-level analysis, all 52 African countries with reliable estimates on the volume of slave exports associated with the four major slave trades are included, except Seychelles, which is excluded due to the lack of data on the share of employment in industrial sectors in the year 2000.

For the individual-level analysis, respondents with missing occupation data, as well as those reporting "never had a job" or "student," are excluded from the analysis. As a result, the sample size is reduced from 53,935 to 40,666 respondents, covering 36 African countries.

3.3 Summary Statistics

Appendix A Table A.1 presents the summary statistics for the country-level variables. Due to the absence of data on the main dependent variable (Share industrial sector_c), Seychelles is excluded from the sample, resulting in an analysis of 51 African countries. The table reveals that the proportion of workers employed in industrial sectors in 2000 is relatively low (12.82%), showing the limited extent of contemporary industrialization in Africa. There is significant variation (a mean of 3.3687 associated with a 3.8526 standard deviation) in the primary independent variable of interest, the volume of the historical slave trade, $\ln(\text{exports/area})_c$, which serves as the foundation of the analysis. In addition, the variation in the four distance variables—Atlantic distance_c, Indian distance_c, Saharan distance_c, and Red Sea distance_c—are also sufficiently large, ensuring the feasibility of my instrumental variable estimations.

Appendix A Table A.2 reports the summary statistics for the variables employed in the individual-level analysis. The data indicates that the probability of a respondent being employed in industrial sectors is relatively low, standing at 21.94% in 2016, which falls significantly short when compared

to those in developed or even similarly developing countries outside Africa. The average and median ages of the respondents range between 35 and 40, as students are excluded from the sample, and the sex ratio is approximately balanced at 1:1. Educational attainment among the respondents is generally low, with the median respondent having only completed primary school (coded as 3 in the Afrobarometer survey). Furthermore, more than half of the respondents reside in rural areas (coded as 0 in the Afrobarometer survey).⁵ The summary statistics of educational attainment and urban or rural residency potentially suggest the low level of industrialization within the sampling countries. Thus, in all individual-level regressions, I control for all these individual characteristics to mitigate potential omitted variable bias. Last but not least, the main independent variable of interest, along with the instrumental variables, exhibits significant variation.

4 Stylized Facts

In this section, I present two stylized graphs: the first displays the correlation between the volume of the historical slave trade and contemporary industrialization, while the second exhibits the relationship between the distances from each country to the primary destinations of the trans-Atlantic slave trade and the volume of the slave trades.

Fact 1. Contemporary Industrialization Is Negatively Correlated with the Volume of the Historical Slave Trade

To explore the relationship between the historical slave trade and contemporary industrialization, a preliminary correlation analysis is conducted between the share of employment in industrial sectors in the year 2000 ($\text{Share industrial sector}_c$) and the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area, $\ln(\text{exports/area})_c$. The correlation plot presented in Figure 1 reveals a clear negative relationship between these two variables, suggesting that countries with higher volume of slave exports tend to exhibit lower contemporary industrial employment shares. The coefficient of the OLS fit is -1.0191 and is statistically significant at 1% level, as in the regression model in Column (1) of Subsection 6.1 Table 1, where no control variables are included.

⁵See more coding details of the Afrobarometer survey in Appendix C.

Notably, the observed negative correlation between the volume of the historical slave trade and contemporary industrialization is not driven by few outliers but rather reflects a general pattern across most African countries. Motivated by this correlation, naturally, the next step is to rigorously assess whether this observed negative correlation is causal, which will be systematically tested in the subsequent sections.

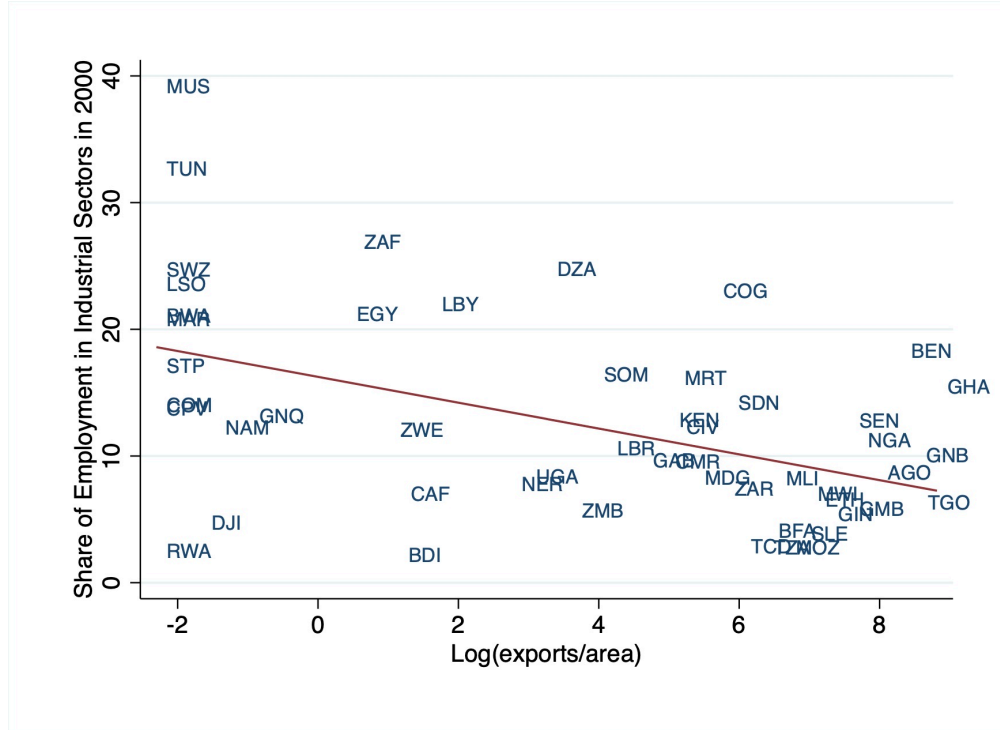


Figure 1: Contemporary Share of Employment in Industrial Sectors and the Volume of the Historical Slave Trade

Notes: The figure displays the correlation between the share of employment in industrial sectors in 2000 ($\text{Share industrial sector}_c$) and the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$).

Fact 2. The Volume of the Slave Trades is Negatively Correlated with the Distances from Each Country to the Primary Destinations of the Trans-Atlantic Slave Trade

The second stylized graph illustrates the relevance condition of the instrumental variables, the distances from each country to the primary destinations of the four major slave trades. As noted in Nunn (2008) and my historical background Subsection 2.1, the trans-Atlantic slave trade had the most significant impact among the four historical slave trades. Thus, I present the correlation be-

tween the volume of the historical slave trade, $\ln(\text{exports/area})_c$, and the distances from each country to the primary destinations (i.e., the West Indies and the southern United States) of the trans-Atlantic slave trade (Atlantic distance_c). Figure 2 reveals a negative relationship between the two variables, suggesting that the volume of the historical slave trade was largely driven by a country's proximity to the primary destinations of the trans-Atlantic slave trade: the closer the country to the destinations, the greater the number of slaves exported. This evidence reinforces the validity of using distances from each country to the primary destinations of major slave trades as instrumental variables. Note that while this graph demonstrates the relevance condition, readers should refer to Subsubsection 6.1.4 for a detailed discussion of the exclusion restriction condition for the instrumental variables.

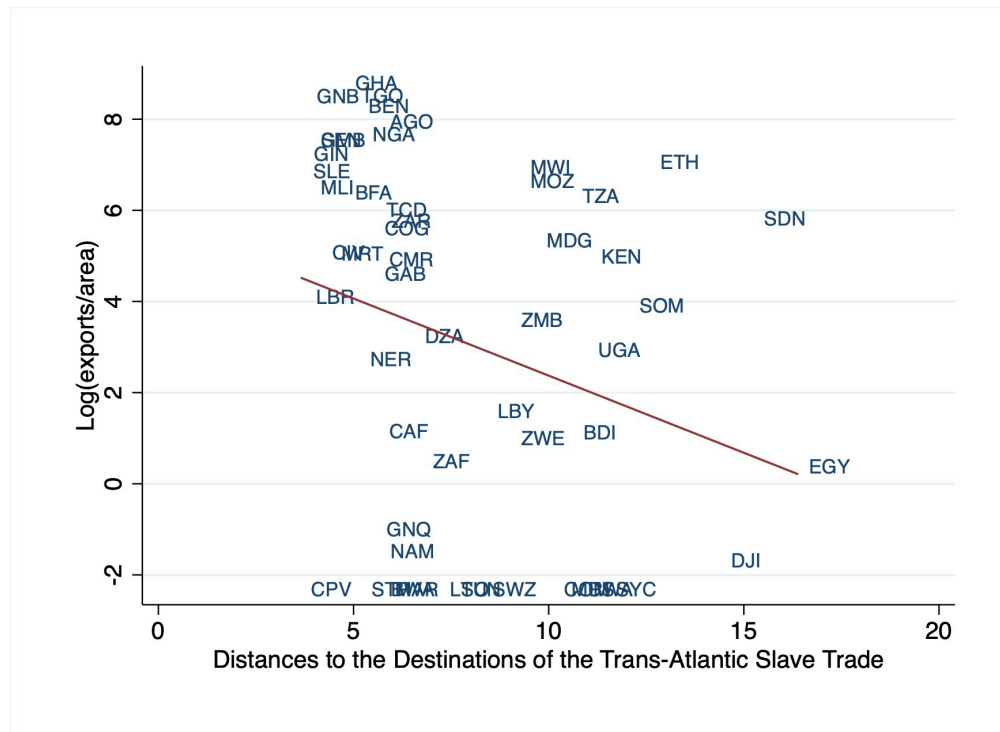


Figure 2: The Volume of the Historical Slave Trade and Distances from each Country to the Primary Destinations of the Trans-Atlantic Slave Trade

Notes: The figure displays the correlation between the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$) and the distances from each country to the primary destinations (i.e., the West Indies and the southern United States) of the trans-Atlantic slave trade (Atlantic distance_c).

In Appendix B, Figures B.1, B.2, and B.3 present correlation analyses analogous to Figure 2 for the trans-Saharan, Indian Ocean, and Red Sea slave trades. Interestingly, The results reveal a negative

correlation only for the trans-Saharan slave trade, highlighting the dominance of the trans-Atlantic slave trade in the overall volume of the historical slave trade, consistent with Nunn (2008) and the historical background provided in Subsection 2.1. Nonetheless, the absence of a negative correlation between the volume of the historical slave trade and the distances from each country to the primary destinations of the Indian Ocean and Red Sea slave trades⁶ does not undermine the validity of these two distances as instrumental variables. When the distances from each country to the primary destinations of each of the four slave trades are included simultaneously, they collectively exhibit significantly negative coefficients at various specifications in the first-stage regressions, as will be demonstrated in Tables 2 and 4.

5 Empirical Strategy

In this section, I outline the empirical strategy at both the country level and the individual level. The various control variables are also described in detail as they are introduced.

5.1 Country-level Analysis

To formally assess the negative correlation between the volume of the historical slave trade and contemporary industrialization in the Stylized Facts Section 4 Figure 1, I first estimate the following country-level equation:

$$\text{Share industrial sector}_c = \beta_0 + \beta_1 \ln(\text{exports/area})_c + \text{colony}'_c \delta + \text{geo}'_c \gamma + \text{climate}'_c \Phi + \varepsilon_c \quad (1)$$

where $\text{Share industrial sector}_c$ represents the proportion of employment in industrial sectors in country c as of the year 2000. $\ln(\text{exports/area})_c$ denotes the natural log of the number of slaves exported from country c between 1400 and 1900 during the four major slave trades, normalized by land area.

The vector colony'_c includes a set of dummy variables indicating the colonial power prior to country c 's independence. The vector geo'_c encompasses geographical controls, including absolute latitude (distance from the equator), longitude, and proximity to the ocean (measured by the natural

⁶There is in fact a slightly positive correlation, though it is negligible due to a lack of statistical significance.

log of coastline divided by land area). The vector $\text{climate}'_c$ includes climatic variables, specifically, minimum monthly rainfall, maximum monthly afternoon average humidity, and the minimum of the average monthly lowest temperature. Most of these geographical and climatic variables are intended to account for whether a country has a tropical climate, which influences the prevalence of infectious diseases and affects agricultural productivity (Kamarck, 1976; Sachs et al., 2001). The inclusion of them also rules out potential arguments that geographical differences impacted industrialization based on the "geographical hypothesis" of economic growth (Sachs, 2000, 2001; Lewis, 2009). At this level of analysis, heteroskedasticity-robust standard errors are employed.

In addition to incorporating control variables, to more rigorously assess whether the observed negative relationship between the historical slave trade and contemporary industrialization is causal, I instrument $\ln(\text{exports/area})_c$ with the distances from each country to the primary destinations of the slave trades, following the methodology established by Nunn (2008). In the Stylized Facts Section 4 Figure 2, I demonstrate the relevance condition of the instrumental variables, while a detailed discussion on the exclusion restriction condition of the instrumental variables is provided in Subsubsection 6.1.4.

5.2 Individual-level Analysis

The above country-level analysis is inevitably constrained by the limited number of observations. To strengthen my results and account for worker-level heterogeneity, I employ the following individual-level regression:

$$\mathbb{1}\{\text{Industrial job}\}_i = \alpha_0 + \alpha_1 \ln(\text{exports/area})_c + \mathbf{X}'_i \theta + \text{colony}'_c \delta + \text{geo}'_c \gamma + \text{climate}'_c \Phi + \epsilon_i \quad (2)$$

where $\mathbb{1}\{\text{Industrial job}\}_i$ is a binary variable indicating whether individual i works in an industrial job at the time of survey (2016). $\mathbf{X}'_i$ is a vector of individual-level controls, including the respondent's age, age squared, gender, urban or rural residency, education level, living conditions, and religion.⁷ Once again, to address potential endogeneity concerns, I include colonial, geographical, and climatic controls. $\ln(\text{export/area})_c$ will also be instrumented using the distances from each country to the

⁷See Appendix C for detailed explanation of these variables.

primary destinations of the slave trades. Standard errors here are clustered at country level.

6 Main Results

In this section, I present the estimation results of the empirical strategies outlined in Section 5. I provide evidence of a robust causal adverse effect of the historical slave trade on contemporary industrialization, supported by rigorous controls and an instrumental variable approach. The validity of the instruments is also discussed.

6.1 Country-level Results

6.1.1 OLS Results

The OLS results for Equation (1) are reported in Table 1. Column (1) presents the model without any control variables, which mirror the OLS fit depicted in the Stylized Facts Section 4 Figure 1. In Column (2), colonizer dummies are added to account for differences in colonial history. Column (3) introduces geographical and climatic controls. As an additional robustness check, Column (4) excludes island nations and North African countries to ensure that the results are not driven by these distant regions, which were relatively less affected by the historical slave trade. Across all specifications, the coefficient of interest (β_1) remains consistently significant, demonstrating a robust negative relationship between the contemporary share of employment in industrial sectors and the volume of the historical the slave trade.

Table 1: Country-level OLS Estimation: Contemporary Share of Employment in Industrial Sectors and the Historical Slave Trade

	(1)	(2)	(3)	(4)
	Share industrial sector _c			
$\ln(\text{exports/area})_c$	-1.0191*** (0.3050)	-1.1329*** (0.3133)	-0.7252** (0.3252)	-0.7000* (0.3753)
Implied effect of one standard deviation increase	-3.9262	-4.3645	-2.7937	-2.4478
Implied effect of 25 to 75 percentile movement	-7.7903	-8.6599	-5.5433	-4.0585
Colonizer dummies	-	✓	✓	✓
Geographical controls	-	-	✓	✓
Climatic controls	-	-	✓	✓
N	51	51	51	42
R^2	0.2282	0.3955	0.5479	0.4131

Notes: The table reports OLS estimates of Equation (1). The dependent variable is the share of employment in industrial sectors in 2000 (Share industrial sector_c). The independent variable is the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$). Column (4) excludes island nations and North African countries. Country-level sample is used. Robust standard errors are displayed in parentheses below each coefficient. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

6.1.2 Selection on Unobservables

Despite the inclusion of a wide range of observable controls, unobservable factors might still bias my estimates. To evaluate the potential impact of such unobservable factors on my estimates, I apply the methodology proposed by [Altonji et al. \(2005\)](#). Their approach calculates the extent to which the effects of unobservable factors would need to exceed that of observable factors in order to fully explain away the results. To introduce their method, consider the estimated coefficient of interest (β_1) in column (1), where no controls are included; let this be denoted as β_1^1 . Similarly, denote the estimated coefficient in column (3), where all available controls are included and no country is dropped, as β_1^3 . The ratio of interest is then calculated as $\frac{\beta_1^3}{\beta_1^1 - \beta_1^3}$.⁸

To understand the intuition behind this ratio, it is helpful to decompose it into the numerator

⁸For further details on the derivation and interpretation of this measure, see [Altonji et al. \(2005\)](#).

and the denominator. First, the measure increases as the denominator ($\beta_1^1 - \beta_1^3$) decreases. This implies that when the difference between β_1^1 and β_1^3 is small, the estimate is less affected by selection on observables, meaning the selection on unobservables would need to be substantially stronger, relative to observables, to fully explain away the estimated effect. Second, the measure is increasing in the numerator (β_1^3), which is intuitive because the larger the value of β_1^3 , the more significant is the effect that needs to be explained by selection on unobservables.

Using this method, calculation yields that selection on unobservable factors would need to be approximately 2.47 ($\frac{-0.7252}{-1.0191 - (-0.7252)}$) times stronger than selection on observable factors to fully explain away the whole OLS estimates. Given that I already account for a wide range of observable variables, as detailed in the Empirical Strategy Subsection 5.1, this degree of selection on unobservables is unlikely. Thus, the OLS estimates are unlikely to suffer from substantial bias.

6.1.3 IV Results

Although it seems that my OLS estimates are not substantially biased, I still try to further address potential selection on unobservables and reverse causality by implementing an instrumental variable approach, as outlined in the Empirical Strategy Subsection 5.1. To elaborate, I instrument the main independent variable of interest, $\ln(\text{exports/area})_c$, using the distances from each country to the primary destinations in each of the four major slave trades. The results of this instrumental variable analysis are presented in Table 2. The specifications in the four columns parallel those in Table 1, with the key difference being the use of instrumental variables for $\ln(\text{exports/area})_c$. The coefficient of interest (β_1) remains significantly negative across all columns. The magnitude of these coefficients is larger than OLS estimates, which is consistent with my expectation. This increase in magnitude may be attributed to potential upward bias arising from the unobservable selective involvement of certain types of societies in the historical slave trade, which underscores the value of the instrumental variable strategy, as it effectively mitigates potential selection on unobservable factors that cannot be directly controlled for. Additionally, the attenuation effect caused by measurement error in the estimation of the volume of the historical slave trade could also contribute to the increase in magnitude.

In the first-stage results, the coefficients for the distances from each country to the primary des-

tinations of the four major slave trades are predominantly significantly negative,⁹ which aligns with the intuitive expectation that countries located farther from the destinations were less likely to be targeted by the slave trades, as discussed in the second part of the Stylized Facts Section 4. The only concern here is that the F-statistics on excluded instruments are generally low, which potentially suggests a weak instrument problem. However, my F-statistics on excluded instruments are close to those in Nunn (2008). Furthermore, these F-statistics on excluded instruments increase in the individual-level analysis, specifically, in Subsection 6.2 Table 4. In summary, the instrumental variable estimates corroborate the OLS findings, establishing a causal negative relationship between the volume of the historical slave trade and the contemporary share of employment in industrial sectors.

Quantitatively, based on the baseline specification in Column (3) of Table 2, a one standard deviation increase in the volume of the historical slave trade corresponds to a 8.75% decrease in the contemporary share of employment in industrial sectors. Alternatively, shifting a country from the 25th percentile to the 75th percentile in the volume of the historical slave trade, while holding all other factors constant, results in a 17.36% decrease in the contemporary share of employment in industrial sectors. These figures are notably large, considering that the mean contemporary share of employment in industrial sectors as of the year 2000 within my country-level sample is merely 12.82%. This suggests that the impact of the historical slave trade on contemporary industrialization is both statistically significant and economically meaningful.

⁹The coefficients on the distances from each country to the primary destinations of the Red Sea slave trade are insignificant, but this is similar to the estimation results in Table IV of Nunn (2008), suggesting that the influence of the Red Sea slave trade is relatively minor among the four major slave trades.

Table 2: Country-level IV Estimation: Contemporary Share of Employment in Industrial Sectors and the Historical Slave Trade

	(1)	(2)	(3)	(4)
	Share industrial sector _c			
$\ln(\text{exports/area})_c$	-2.1518*** (0.5984)	-2.0699*** (0.5012)	-2.2703* (1.1655)	-1.5867** (0.6546)
Implied effect of one standard deviation increase	-8.2901	-7.9744	-8.7467	-5.5481
Implied effect of 25 to 75 percentile movement	-16.4491	-15.8226	-17.3550	-9.1991
	First Stage. Dependent Variable: $\ln(\text{exports/area})_c$			
Atlantic distance _c	-1.2787*** (0.3591)	-1.6955*** (0.4343)	-1.3131 (0.7779)	-1.6856** (0.6798)
Indian distance _c	-1.0789*** (0.3806)	-1.4132** (0.5355)	-1.0776 (0.7148)	-1.5704* (0.8007)
Saharan distance _c	-2.3720*** (0.8267)	-2.9295*** (1.0639)	-1.1447 (1.6267)	-4.0756** (1.5518)
Red Sea distance _c	-0.0142 (0.7113)	-0.1362 (0.8195)	-1.2038 (1.9209)	2.1297 (2.4038)
F-stat of excluded instruments	4.10	4.32	0.76	2.86
Colonizer dummies	-	✓	✓	✓
Geographical controls	-	-	✓	✓
Climatic controls	-	-	✓	✓
<i>N</i>	51	51	51	42

Notes: The table reports IV estimates of Equation (1). The dependent variable is the share of employment in industrial sectors in 2000 (Share industrial sector_c). The independent variable is the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$). Column (4) excludes island nations and North African countries. The excluded instruments are the distances from each country to the primary destinations in each of the four major slave trades. Country-level sample is used. Robust standard errors are displayed in parentheses below each coefficient. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

6.1.4 Exclusion Restriction

My instrumental variable strategy rests on the assumption that the distances from each country to the primary destinations of the slave trades affect contemporary industrialization only through

the slave trade channel. A primary concern is that these distance variables may be correlated with whether a country has a coastline, since coastal access could positively affect industrialization given that port cities generally develop earlier. To address this, I control for proximity to the ocean by including the ratio of coastline to land area in the first-stage regression, thereby mitigating this concern. Furthermore, in my dataset, the distance variables are nearly uncorrelated with proximity to the ocean,¹⁰ suggesting that this channel is unlikely to bias my estimates. More broadly, as discussed in Nunn (2008), the scale of the slave trades was unprecedented. Consequently, pre-existing international trade routes and ports had limited influence in determining the locations from which slaves were exported.

6.2 Individual-level Results

6.2.1 OLS Results

To estimate the impact of the historical slave trade on contemporary industrialization at the individual level, I implement Equation (2) and present the results in Table 3. Column (1) provides estimates without any controls. Individual-level controls are introduced in Column (2), followed by the inclusion of colonial history controls in Column (3). Geographical and climatic variables are further incorporated in Column (4). Across all specifications, the coefficient of interest (α_1) remains consistently negative and statistically significant, mirroring the negative relationship between contemporary industrialization and the historical slave trade observed in the country-level analysis shown in Table 1.

¹⁰The absolute values of the correlation coefficients between proximity to the ocean and the four distance variables are all below 0.1.

Table 3: Individual-level OLS Estimation: Indicator Variable of Industrial Job and the Historical Slave Trade

	(1)	(2)	(3)	(4)
	$\mathbb{1}\{\text{Industrial job}\}_i$			
$\ln(\text{exports/area})_c$	-0.0186*** (0.0055)	-0.0181*** (0.0051)	-0.0206*** (0.0046)	-0.0152*** (0.0043)
Implied effect of one standard deviation increase	-0.0688	-0.0670	-0.0765	-0.0564
Implied effect of 25 to 75 percentile movement	-0.1141	-0.1110	-0.1268	-0.0936
Individual controls	-	✓	✓	✓
Colonizer dummies	-	-	✓	✓
Geographical controls	-	-	-	✓
Climatic controls	-	-	-	✓
N	40666	39514	39514	39514
R^2	0.0276	0.1267	0.1398	0.1493

Notes: The table reports OLS estimates of Equation (2). The dependent variable is an indicator variable of whether individual i works in industrial sectors or not ($\mathbb{1}\{\text{Industrial job}\}_i$). The independent variable is the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$). Individual-level sample from Round 6 (2016) Afrobarometer survey is used. Standard errors are clustered at country level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

6.2.2 Selection on Unobservables

Using the methodology described in Subsubsection 6.1.2, I estimate the extent to which the influence of unobservable factors would need to exceed that of observable factors to fully explain away the OLS estimates. Specifically, I apply the procedure to the coefficients on the variable of interest (α_1) in Column (1) and Column (4) of Table 3, denoted as α_1^1 and α_1^4 , respectively, and then compute the ratio $\frac{\alpha_1^4}{\alpha_1^1 - \alpha_1^4}$. In this case, selection on unobservables would have to be approximately 4.47 ($\frac{-0.0152}{-0.0186 - (-0.0152)}$) times greater than selection on observables to nullify the whole OLS estimates, indicating that the possibility of substantial bias in the individual-level OLS estimates is even lower than at the country level. This further strengthens the reliability of my OLS results.

6.2.3 IV Results

Nevertheless, I still employ the instrumental variable strategy in the individual-level analysis. The estimations, presented in Table 4, follow the same specifications as those in Table 3 except the use of the instrumental variables. The instrumental variable estimation yields slightly more negative coefficients, consistent with the potential selection on unobservables and measurement error issues previously discussed. To sum up, the instrumental variable estimation at the individual level reinforces the causal negative relationship between the historical slave trade and contemporary industrialization observed at the country level in Table 2.

According to the baseline specification in Column (4) of Table 4, a one standard deviation increase in the volume of the historical slave trade corresponds to a 5.94% decrease in the probability of a respondent being employed in industrial sectors. Alternatively, transitioning a respondent from a country in the 25th percentile to one in the 75th percentile in terms of the volume of the historical slave trade, while holding all other variables constant, results in a 9.84% decrease in the probability of that respondent being employed in industrial sectors. These figures are economically significant, given that the average probability of being employed in industrial sectors for a individual-level sample respondent is only 21.94% from summary statistics in Appendix A Table A.2. This underscores that the long-term impact of the historical slave trade on contemporary industrialization is also evident at the individual level using microdata.

Table 4: Individual-level IV Estimation: Indicator Variable of Industrial Job and the Historical Slave Trade

	(1)	(2)	(3)	(4)
	$\mathbb{1}\{\text{Industrial job}\}_i$			
$\ln(\text{exports/area})_c$	-0.0323*** (0.0090)	-0.0336*** (0.0079)	-0.0347*** (0.0079)	-0.0160** (0.0064)
Implied effect of one standard deviation increase	-0.1200	-0.1246	-0.1289	-0.0594
Implied effect of 25 to 75 percentile movement	-0.1989	-0.2065	-0.2137	-0.0984
First Stage. Dependent Variable: $\ln(\text{exports/area})_c$				
Atlantic distance _c	-1.7939*** (0.4255)	-1.6295*** (0.4861)	-1.5549*** (0.4833)	-0.9696 (0.7209)
Indian distance _c	-1.6668*** (0.4468)	-1.6409*** (0.4427)	-1.6230*** (0.4908)	-1.3273* (0.7413)
Saharan distance _c	-3.1466*** (0.9079)	-3.0581*** (1.0191)	-3.0557*** (1.1770)	1.1155 (1.7356)
Red Sea distance _c	-0.3533 (0.8864)	-0.2346 (1.1250)	-0.1084 (1.2147)	-5.5374*** (1.7851)
F-stat of excluded instruments	5.14	5.10	4.06	3.56
Individual controls	-	✓	✓	✓
Colonizer dummies	-	-	✓	✓
Geographical controls	-	-	-	✓
Climatic controls	-	-	-	✓
<i>N</i>	40666	39514	39514	39514

Notes: The table reports IV estimates of Equation (2). The dependent variable is an indicator variable of whether individual i works in industrial sectors or not ($\mathbb{1}\{\text{Industrial job}\}_i$). The independent variable is the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$). The excluded instruments are the distances from each country to the primary destinations in each of the four major slave trades. Individual-level sample from Round 6 (2016) Afrobarometer survey is used. Standard errors are clustered at country level. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

7 Mechanisms of Persistence

In this section, I examine how the adverse impact of the historical slave trade on contemporary industrialization persisted over centuries. In particular, I explore how institutional factors sustained and even exacerbated the detrimental impact of the historical slave trade on the process of industrialization in Africa.

7.1 Background: Poor Institutional Quality in Africa

Among the potential factors—such as capital accumulation, infrastructure development, and institutional quality—that may have perpetuated the impact of the historical slave trade on industrialization, I focus on institutional quality, as institutions that promote investments play a crucial role in fostering industrialization ([Acemoglu et al., 2002](#)). Furthermore, early investments from institutions are critical for long-term development. For instance, the provision of public goods ([Huillery, 2009](#); [Cagé and Rueda, 2016](#); [Jedwab and Moradi, 2016](#)), the establishment of schools ([Cogneau and Moradi, 2014](#); [Wantchekon et al., 2015](#)), and even temporary subsidies ([Ehrlich and Seidel, 2018](#)) implemented by historical institutions have played a substantial role in shaping current economic performance. To lay the groundwork for my subsequent analysis related to institutional factors, I briefly review the markedly deficient institutional quality in Africa as follows.

Following the discussion on the early stages of industrialization of African countries in the Historical Background Subsection 2.2, the industrialization process in Africa is frequently marked by significant government intervention. According to [Bates \(2014, 2015\)](#), policies such as import licensing and protective tariffs were often implemented with the (originally good) intention of fostering domestic industrial sectors in African nations. Additionally, the production rights for specific industrial goods were subject to strict government regulation. However, these interventions often prioritized political connections over economic efficiency in determining which sectors received protection and which firms were granted production rights. In practice, authoritarian regimes in Africa frequently imposed policies that favored their political allies, thereby consolidating their ruling power. As a result, the post-independence industrial sectors in Africa became highly monopolized and inefficient, characterized by large but redundant production facilities, which significantly hindered its contemporary industrialization process.

7.2 Empirical Challenges

Given the crucial role that governments played in shaping the trajectory of contemporary industrialization in Africa and the well-established fact that better institutions facilitate industrialization ([Acemoglu et al., 2002](#)), it seems intuitive to investigate how the adverse impact of the historical slave trade on industrialization varies across countries with differing levels of institutional quality. However, current institutional quality may be endogenous, potentially resulting in biased estimates. For example, if current institutional quality is correlated with colonial or post-colonial developments ([Acemoglu et al., 2001b](#)), then observing a stronger negative impact of the slave trades on contemporary industrialization in countries with lower institutional quality today may conflate the effects of the slave trades with those of colonial or post-colonial developments.

Fortunately, as marked in Subsection 2.3, [Gennaioli and Rainer \(2007\)](#) provides evidence that the degree of pre-colonial institutional centralization significantly influences current institutional quality. Centralization of pre-colonial institutions facilitated the provision of public goods by making local chiefs more accountable during the colonial and post-colonial periods. Most importantly, the degree of pre-colonial institutional centralization is associated with current institutional quality but, unlike the latter, it is not a consequence of colonial or post-colonial developments, as it pre-dates them. Therefore, by examining whether the adverse impact of the historical slave trade on contemporary industrialization is more pronounced in regions with lower pre-colonial institutional centralization, I can accurately assess whether poorer institutional quality sustained the adverse impact of the historical slave trade on industrialization.

7.3 Results

Building on the rationale discussed above, I first conduct a country-level analysis by estimating Equation (1) separately for countries with high pre-colonial institutional centralization and those with low pre-colonial institutional centralization. The results are presented in Panel A of Table 5. Column (1) includes countries characterized by low pre-colonial institutional centralization (petty/paramount chiefdoms), while column (2) includes the remaining half, consisting of countries marked by high pre-colonial institutional centralization (large states). Unfortunately, due to the extremely small sample sizes, the estimated coefficients in both samples are statistically insignificant.

Consequently, no reliable inferences can be drawn from these imprecisely estimated coefficients, which motivates the adoption of individual-level analysis.¹¹

An individual-level analysis estimating Equation (2) separately for respondents residing in countries with high pre-colonial institutional centralization and those residing in countries with low pre-colonial institutional centralization is presented in Panel B of Table 5. The estimated coefficient of interest (α_1) for respondents living in countries with high pre-colonial institutional centralization (-0.014) is considerably smaller in magnitude than that for respondents living in countries with low pre-colonial institutional centralization (-0.0489). Both coefficients are at least marginally significant, with the coefficient for respondents in countries of low pre-colonial institutional centralization being significant at the 1% level. Furthermore, the estimated coefficient for respondents in countries with high pre-colonial institutional centralization in the instrumental variable specification analogous to column (4) of Subsection 6.2 Table 4 is statistically insignificant (-0.0041 , with a standard error of 0.0279).¹² This suggests that the historical slave trade has significantly less influence on contemporary industrialization in regions characterized by high pre-colonial institutional centralization, where the subsequent institutional quality was also expected to be better (Gennaioli and Rainer, 2007).

Given the evidence presented, I propose an institutional explanation to how the historical slave trade persistently influenced contemporary industrialization: The absence of reliable institutions exacerbated the difficulties faced by African countries raided by the historical slave trade in developing industrial sectors, as industrialization in these countries was often impeded by dysfunctional or even discriminatory governance. Therefore, poor institutional quality sustained and even exacerbated the adverse impact of the historical slave trade on industrialization.

¹¹Note that I do not employ an instrumental variable strategy in the country-level analysis due to the insufficient number of observations.

¹²The variation in the variable of interest ($\ln(\text{exports/area})_c$) is insufficient in the sample of respondents in countries with low pre-colonial institutional centralization, resulting in collinearity with the instrumental variables. Consequently, I cannot perform the instrumental variable estimation in the sample of respondents in countries with low pre-colonial institutional centralization.

Table 5: Country-level and Individual-level OLS Estimation: Industrialization, the Historical Slave Trade, and Pre-colonial Institutional Centralization

	(1)	(2)
Pre-colonial Institutional Centralization	Low	High
Panel A. Country-level Analysis, Dependent Variable: $\text{Share industrial sector}_c$		
$\ln(\text{exports/area})_c$	2.3921 (1.9238)	-0.3450 (0.3876)
Institutional centralization score	✓	✓
Colonizer dummies	✓	✓
Geographical controls	✓	✓
Climatic controls	✓	✓
N	14	33
R^2	0.888	0.782
Panel B. Country-based Individual-level Analysis, Dependent Variable: $\mathbb{1}\{\text{Industrial job}\}_i$		
$\ln(\text{exports/area})_c$	-0.0489*** (0.0068)	-0.0140* (0.0081)
Implied effect of one standard deviation increase	-0.1213	-0.0507
Implied effect of 25 to 75 percentile movement	-0.1146	-0.0829
Institutional centralization score	✓	✓
Individual controls	✓	✓
Colonizer dummies	✓	✓
Geographical controls	✓	✓
Climatic controls	✓	✓
N	8676	28195
R^2	0.1423	0.1672

Notes: The table reports OLS estimates of Equation (1) and Equation (2) using the sample of countries with high pre-colonial institutional centralization and those with low pre-colonial institutional centralization separately. The dependent variable is the share of employment in industrial sectors in 2000 ($\text{Share industrial sector}_c$) in Panel A, and an indicator variable of whether individual i works in industrial sectors or not ($\mathbb{1}\{\text{Industrial job}\}_i$) in Panel B. The independent variable is the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports/area})_c$) in both panels. Country-level sample is used in Panel A. Individual-level sample from Round 6 (2016) Afrobarometer survey is used in Panel B. Robust standard errors are displayed in Panel A. Standard errors are clustered at country level in Panel B. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Quantitatively, according to the estimates in Panel B of Table 5, a one standard deviation increase in the volume of the historical slave trade corresponds to a 5.07% decrease in the probability of a respondent being employed in industrial sectors in countries with high pre-colonial institutional centralization, while in countries with low pre-colonial institutional centralization, the analogous implied effect is nearly 2.5 times larger. A similar difference can be observed when comparing respondents from a country in the 25th percentile to one in the 75th percentile in terms of the volume of the historical slave trade, although the difference is slightly smaller due to the larger interquartile range in countries with high pre-colonial institutional centralization.

It is important to note that my finding—that the adverse impact of the historical slave trade is sustained by an external factor, institutional quality—should be viewed as complementary rather than contradictory to the internal transmission of the slave trades’ effects on trust through norms and beliefs, as discussed in [Nunn and Wantchekon \(2011\)](#). It is reasonable to expect that tangible legacies, such as industrialization, are sustained by external factors like institutional quality, whereas intangible consequences, such as trust, are transmitted through internal factors like norms and beliefs. The historical slave trade shapes different dimensions of African countries today through different channels, and my analysis of the external transmission of its legacies contributes to a more comprehensive understanding of these long-term effects.

In recent literature on the persistence of historical events, the interaction between institutional and cultural forces has received considerable emphasis. Theoretically, [Bisin and Verdier \(2024\)](#) demonstrates that institutions and culture can function as substitutes or complements, depending on the historical context, in shaping the trajectory of long-run economic development. Earlier studies, such as [Tabellini \(2008\)](#) and [Persson and Tabellini \(2021\)](#), likewise underscore the significance of this interplay between institutions and culture. Empirically, on the one hand, [Guiso et al. \(2004\)](#) shows that individuals residing in areas with stronger institutions rely less on social capital in Italy, providing evidence of substitution between institutions and culture. Similarly, in South Asia, [Jha \(2013\)](#) demonstrates that local cultural traits play a pivotal role in reducing interethnic riots when state institutional power is weak. In Africa, [Lowes et al. \(2017\)](#) and [Lowes and Montero \(2021\)](#) document that when state institutions are more developed, pro-social cultural norms tend to be weaker, and vice versa, further illustrating substitution. On the other hand, [Dell \(2010\)](#) presents evidence

from Peru and Bolivia that landlords with secure property rights (institutions) had greater incentives to invest in public goods (a pro-social norm), pointing to complementarity.

My study, together with [Nunn and Wantchekon \(2011\)](#), suggests that both high institutional quality that fosters economic growth and pro-social cultural norms (specifically trust) are absent, potentially indicating complementarity between institutions and culture. Further research is needed to confirm the relationship between the institutional and cultural legacies of the historical slave trade and to precisely identify the relative contributions of each channel to the persistent effects of the slave trades.

8 Conclusion

In this study, I demonstrate a negative relationship between the historical slave trade and contemporary industrialization in Africa. This relationship is shown to be causal, as it remains robust after accounting for various control variables and applying an instrumental variable strategy. Regarding the persistence of the impact, by analyzing the differential impacts of the historical slave trade across countries with differing degrees of pre-colonial institutional centralization, I find that the adverse impact of the historical slave trade on contemporary industrialization was sustained and even exacerbated by the generally weak institutional quality of African governments.

Unlike previous studies that explore various long-term cultural or health-related outcomes of the historical slave trade, my findings highlight the adverse impact of the historical slave trade on contemporary industrialization as a significant channel through which it contributes to Africa's current economic underperformance (Nunn, 2008). Furthermore, the persistence of this impact through external factors, specifically through institutional quality, complements the emphasis on internal norms and beliefs as key mechanisms for transmitting the effects of historical shocks noted in Nunn and Wantchekon (2011), Teso (2019), and Zhang et al. (2021). This suggests that both external and internal factors play crucial roles in the transmission of the long-term effects of historical shocks.

For African countries to converge with the rest of the world in terms of income levels, industrialization is definitely necessary (Solow, 1956). Achieving this requires the establishment of stable political institutions that can facilitate industrialization (Acemoglu et al., 2002) and, in turn, promote economic growth (DeLong and Shleifer, 1993; Acemoglu et al., 2001b). My research disentangles the persistent adverse impact of the historical slave trade on contemporary industrialization from an institutional perspective, aiming to provide some motivations for African governments to take practical and informed actions.

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Appendix A Additional Tables

Table A.1: Country-level Summary Statistics

	Obs.	Mean	Std. Dev.	P10	P50	P90
Share industrial sector _c	51	12.8153	8.2187	3.9353	11.3161	23.6408
ln(exports/area) _c	51	3.3687	3.8526	-2.3026	4.6274	7.6908
Latitude (distance from the equator) _c	51	13.7373	9.8629	3	12	29
Longitude _c	51	15.9404	19.6543	-10.9392	17.5414	39.6198
Lowest monthly rainfall _c	51	7.6863	13.7543	0	0	23
Average maximum humidity _c	51	71.549	12.0321	57	73	82
Average minimum temperature _c	51	8.549	7.4197	-1	10	17
ln(coastline/area) _c	51	-0.3794	3.1004	-4.6052	0.2469	2.6584
Atlantic distance _c	51	7.3009	3.2613	3.8888	5.6863	11.0833
Indian distance _c	51	7.036	4.2162	2.3583	8.3668	11.5998
Saharan distance _c	51	3.4892	1.5753	1.7682	3.3531	5.8565
Red Sea distance _c	51	3.4689	1.4714	1.3613	3.5289	5.2588

Notes: The table presents the summary statistics of the variables used in my country-level analysis. The country sample comprises 51 countries in Africa.

Table A.2: Individual-level Summary Statistics

	Obs.	Mean	Std. Dev.	P10	P50	P90
$\mathbb{1}\{\text{Industrial job}\}_i$	40,666	0.2194	0.4139	0	0	1
$\ln(\text{exports/area})_c$	40,666	3.7137	3.7083	-2.3026	4.9991	7.6908
Age_i	40,464	39.2464	14.0869	23	36	60
$\mathbb{1}\{\text{Male}\}_i$	40,666	0.4999	0.5	0	0	1
Urban or rural residency $_i$	40,666	1.1989	1.4627	0	0	3
Education level $_i$	40,573	3.2842	2.2309	0	3	6
Living condition $_i$	40,591	2.7413	1.2676	1	3	4
Latitude (distance from the equator) $_c$	40,666	14.4927	10.2497	2	14	30
Longitude $_c$	40,666	17.304	18.9573	-9.3071	23.8204	37.8576
Lowest monthly rainfall $_c$	40,666	9.0527	12.5317	0	3	25
Average maximum humidity $_c$	40,666	68.9841	12.0232	56	72	82
Average minimum temperature $_c$	40,666	7.2561	7.3476	-2	8	17
$\ln(\text{coastline/area})_c$	40,666	-0.7333	2.9253	-4.6052	0.0684	1.9477
Atlantic distance $_c$	40,666	7.5462	3.1173	3.8975	6.7659	10.9957
Indian distance $_c$	40,666	6.3142	4.1508	2.1832	5.7646	10.7907
Saharan distance $_c$	40,666	3.7145	1.6485	1.7682	3.3589	6.2739
Red Sea distance $_c$	40,666	3.5627	1.2688	1.6499	3.6955	5.2275

Notes: The table presents the summary statistics for the variables utilized in my individual-level analysis. The sample includes 40,666 respondents from 36 African countries based on the Round 6 (2016) Afrobarometer survey.

Appendix B Additional Figures

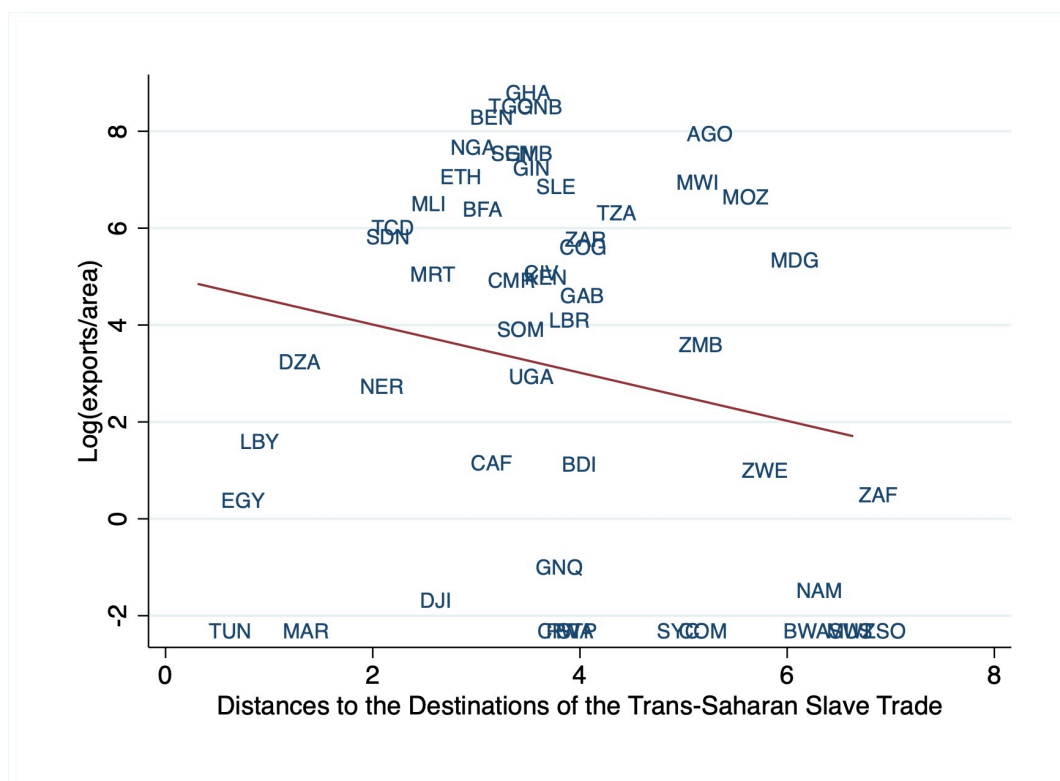


Figure B.1: The Volume of the Historical Slave Trade and Distances from each Country to the Primary Destinations of the Trans-Saharan Slave Trade

Notes: The figure displays the correlation between the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports}/\text{area})_c$) and the distances from each country to the primary destinations of the trans-Saharan slave trade (Saharan distance_c).

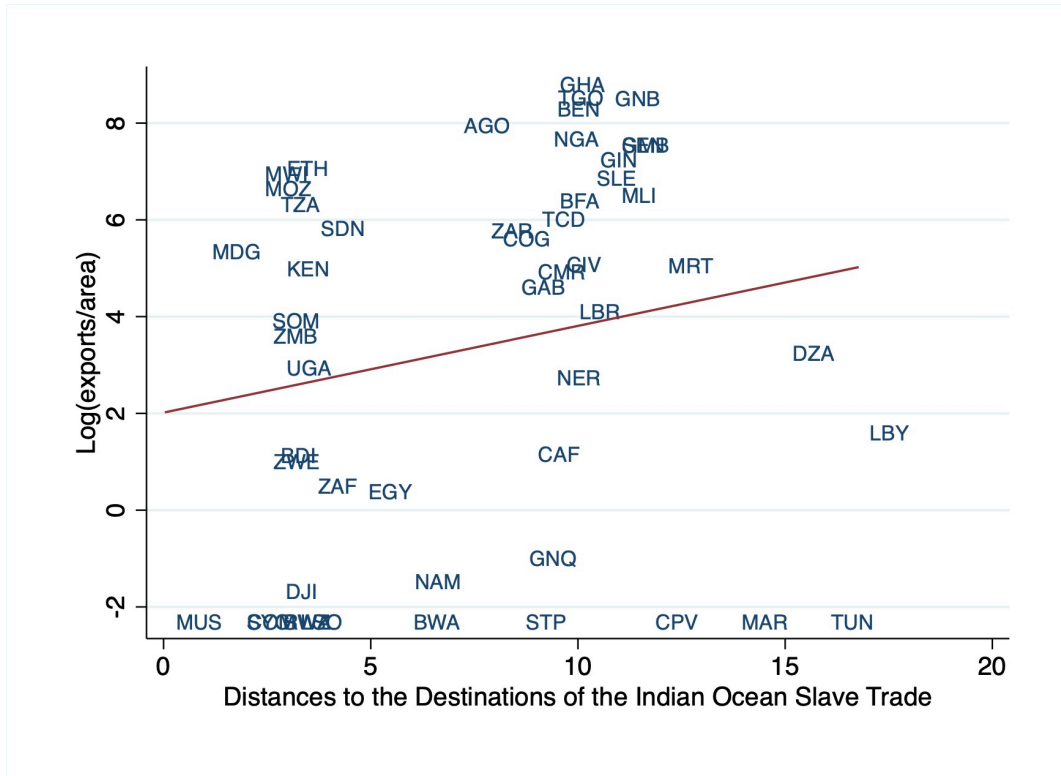


Figure B.2: The Volume of the Historical Slave Trade and Distances from each Country to the Primary Destinations of the Indian Ocean Slave Trade

Notes: The figure displays the correlation between the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports}/\text{area})_c$) and the distances from each country to the primary destinations of the Indian Ocean slave trade (Indian distance_c).

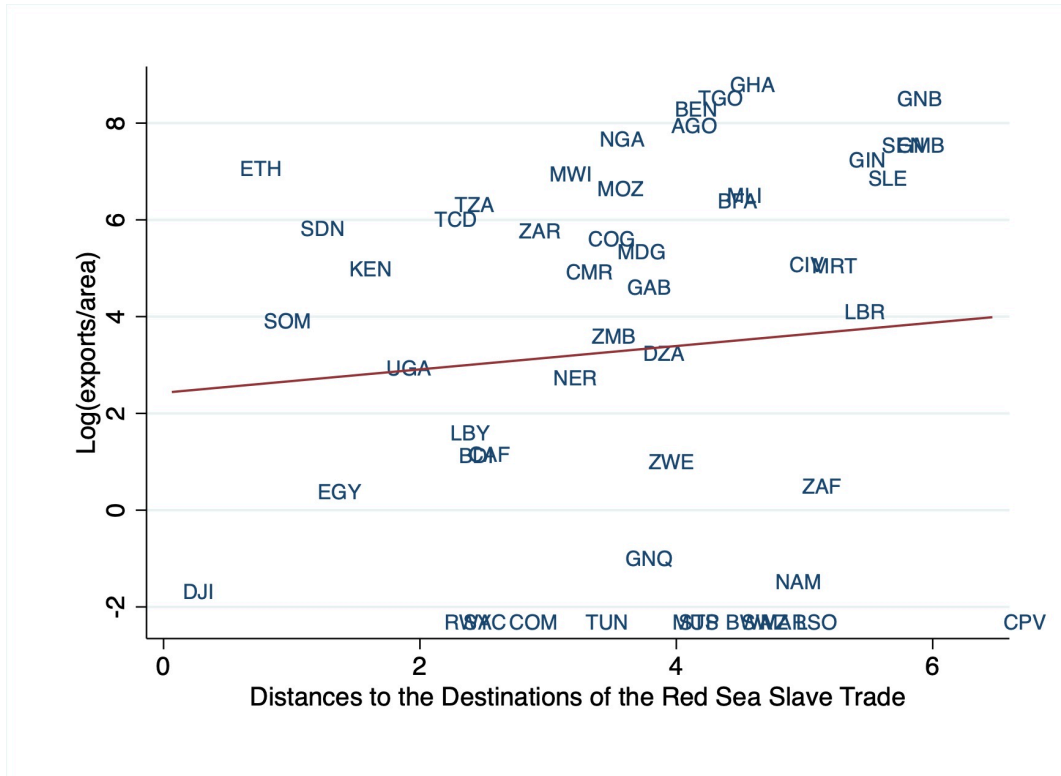


Figure B.3: The Volume of the Historical Slave Trade and Distances from each Country to the Primary Destinations of the Red Sea Slave Trade

Notes: The figure displays the correlation between the natural log of the number of slaves exported from each country between 1400 and 1900 during the four major slave trades, normalized by land area ($\ln(\text{exports}/\text{area})_c$) and the distances from each country to the primary destinations of the Red Sea slave trade ($\text{Red Sea distance}_c$).

Appendix C Detailed Afrobarometer Survey Questions

In this appendix, I present the corresponding Afrobarometer survey questions used to construct the variables in this paper, sourced from the 2016 (Round 6) Afrobarometer survey ([Afrobarometer, 2016](#)). The survey questions and possible answers are obtained from the Afrobarometer [Merged Round 6 Codebook \(2016\)](#).

- Variable: $\mathbb{1}\{\text{Industrial job}\}_i$

Question: What is your main occupation?

Answers: 0=Never had a job, 1=Student, 2=Housewife / homemaker, 3=Agriculture / farming / fishing / forestry, 4=Trader / hawker / vendor, 5=Retail / Shop, 6=Unskilled manual worker (e.g., cleaner, laborer, domestic help, unskilled manufacturing worker), 7=Artisan or skilled manual worker (e.g., trades like electrician, mechanic, machinist or skilled manufacturing worker), 8=Clerical or secretarial, 9=Supervisor / Foreman / Senior Manager, 10=Security services (police, army, private security), 11=Mid-level professional (e.g., teacher, nurse, mid-level government officer), 12=Upper-level professional (e.g., banker/finance, doctor, lawyer, engineer, accountant, professor, senior-level government officer), 95=Other, 99=Don't know, 98=Refused to answer, -1=Missing

- Variable: Age_i

Question: How old are you?

Answers: 18-105, 998=Refused to answer, 999=Don't know, -1=Missing

- Variable: $\mathbb{1}\{\text{Male}\}_i$

Question: Respondent's gender

Answers: 1=Male, 2=Female

- Variable: $\text{Urban or rural residency}_i$

Question: Urban or Rural Residency

Answers: 0=Rural, 1=Peri-urban, 2=Semi-urban, 3=Urban

- Variable: Education level_i

Question: What is your highest level of education?

Answers: 0=No formal schooling, 1=Informal schooling only (including Koranic schooling), 2=Some primary schooling, 3=Primary school completed, 4=Intermediate school or Some secondary school / high school, 5=Secondary school / high school completed, 6=Post-secondary qualifications, other than university e.g. a diploma or degree from a polytechnic or college, 7=Some university, 8=University completed, 9=Post-graduate, 99=Don't know [Do not read], 98=Refused to answer, -1=Missing

► Variable: Religion_i

Question: What is your religion, if any?

Answers: 0=None, 1=Christian only (i.e., respondents says only "Christian", without identifying a specific sub-group), 2=Roman Catholic, 3=Orthodox, 4=Coptic, 5=Anglican, 6=Lutheran, 7=Methodist, 8=Presbyterian, 9=Baptist, 10=Quaker/Friends, 11=Mennonite, 12=Evangelical, 13=Pentecostal (e.g., "Born Again" and/or "Saved"), 14=Independent (e.g., "African Independent Church"), 15=Jehovah's Witness, 16=Seventh Day Adventist, 17=Mormon, 18=Muslim only (i.e., respondents says only "Muslim", without identifying a specific sub-group), 19=Sunni only (i.e., respondents says only "Sunni Muslim", without identifying a specific sub-group), 20=Ismaeli, 21=Mouridiya Brotherhood, 22=Tijaniya Brotherhood, 23=Qadiriya Brotherhood, 24=Shia, 25=Traditional/ethnic religion, 26=Hindu, 27=Bahai, 28=Agnostic (Do not know if there is a God), 29=Atheist (Do not believe in a God), 30=Dutch Reformed, 31=Calvinist, 32=Church of Christ, 33=Zionist Christian Church, 34=Jewish, 35=Assemblies of God Church, 36=New Apostolic, 99=NOT ASKED IN THIS COUNTRY, 100=Celestial Christianity, 220=Christian Rationalism, 260=Apostolic Church, 300=African Inland Church, 420=Apokalypsy, 421=FPVM, 422=Shine, 460=Last Church of God, 461=African International, 462=African Abraham Church, 463=New Apostolic Church, 500=Hamadiya Brotherhood, 501=Wahhabiya Brotherhood, 502=Hamalite / Chérif de Nioro Brotherhood, 503=Ansardine Brotherhood, 540=Old Apostolic, 541=Nazaren Church, 620=Izala, 660=Layene, 820=United Church of Zamia, 822=Christian Missions in Many Lands, 860=Salvation Army, 900=Tamil, 901=Telegu, 902=Marathi, 930=Bashariya Mission, 931=Hisbulah Mission, 1260=Alliance Chrétienne et Missionnaire, 9995=Other, 9998=Refused to answer, 9999=Don't know, -1=Missing

- Variable: Country_i (for merging with $\ln(\text{exports/area})_c$)

Question: Country

Answers: 1=Algeria, 2=Benin, 3=Botswana, 4=Burkina Faso, 5=Burundi, 6=Cameroon, 7=Cape Verde, 8=Cote d'Ivoire, 9=Egypt, 10=Gabon, 11=Ghana, 12=Guinea, 13=Kenya, 14=Lesotho, 15=Liberia, 16=Madagascar, 17=Malawi, 18=Mali, 19=Mauritius, 20=Morocco, 21=Mozambique, 22=Namibia, 23=Niger, 24=Nigeria, 25=São Tomé and Príncipe, 26=Senegal, 27=Sierra Leone, 28=South Africa, 29=Sudan, 30=Swaziland, 31=Tanzania, 32=Togo, 33=Tunisia, 34=Uganda, 35=Zambia, 36=Zimbabwe